WORKED SOLUTION

STEP III 1987, Question 4

Pure • Volumes of Revolution

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Read the question before you go on. It is in the STEP Support Programme's question database:

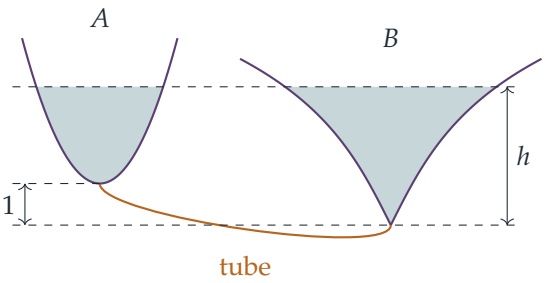
step.maths.org/questions/87-s3-q4

The key idea

The water just moves from one funnel to the other, so its volume is conserved. Each funnel is a solid of revolution about the y-axis, so the volume of water up to a depth d is $\int_0^d \pi x^2 \, \mathrm{d}y$, with x^2 written in terms of y. Once both volumes are known as functions of depth, the only remaining care is with the two different reference levels: the depths in A and B differ by one unit.

Solution

We treat the tube as having negligible volume, and we read the second "closed" in the question as "opened" (otherwise no water could move). Measure y upwards from the bottom of each funnel.



Volumes as functions of depth

For A, the curve is $y = x^2$, so the horizontal cross-section at height y is a disc of area $\pi x^2 = \pi y$. Water of depth d in A has volume

$$V_A(d) = \int_0^d \pi y \, \mathrm{d}y = \frac{\pi d^2}{2}.$$

For B, the curve $y = 2\sinh^{-1} x$ with $x > 0$ inverts to $x = \sinh\frac{y}{2}$. Using $\sinh^2\frac{y}{2} = \frac{1}{2}(\cosh y - 1)$, water of depth d in B has volume

$$V_B(d) = \int_0^d \pi \sinh^2\frac{y}{2}\,\mathrm{d}y = \frac{\pi}{2}\int_0^d (\cosh y - 1)\,\mathrm{d}y = \frac{\pi}{2}(\sinh d - d).$$

Conservation of water

At the start, A holds water of depth 4 and B is empty, so the total volume is

$$V_A(4) = \frac{16\pi}{2} = 8\pi.$$

At the end both surfaces are at height h above the bottom of B. That makes the depth in B equal to h. The bottom of A is one unit higher than the bottom of B, so the depth in A is $h - 1$.

Ask yourself Why do the two surfaces end up at the same level at all?

The two funnels and the tube form one connected body of water at rest. Its pressure depends only on the height, and both free surfaces are at atmospheric pressure, so they are at the same height. The question gives us this, but it explains why the two depths differ by exactly the difference in the heights of the bottoms.

The water was neither gained nor lost, so

$$\frac{\pi(h-1)^2}{2} + \frac{\pi}{2}(\sinh h - h) = 8\pi.$$

Multiplying by $2/\pi$ and expanding,

$$\begin{aligned}(h-1)^2 + \sinh h - h &= 16\\ h^2 - 2h + 1 + \sinh h - h &= 16\\ h^2 - 3h + \sinh h &= 15,\end{aligned}$$

which is the required equation.

The left-hand side of the middle line is a sensible model only for $h \geq 1$, where A actually contains water. On that range its derivative $2(h-1) + \cosh h - 1$ is positive, so the equation has exactly one relevant root. Numerically $h \approx 3.33$, so the water in A drops from depth 4 to about 2.33.

Checks

- Bisection gives $h \approx 3.3272$. Then $V_A(h-1) + V_B(h) \approx 25.1327$, which matches $8\pi \approx 25.1327$.
- Simpson's rule on $\int_0^d \pi\sinh^2(y/2)\,\mathrm{d}y$ agrees with $\frac{\pi}{2}(\sinh d - d)$ for several values of d (for example $d = 1.7$ gives 1.48540 both ways).
- Plausibility: $h - 1 < 4$, so the level in A has fallen, as it must when water flows out of it.

Common slips

- Using depth h in both funnels, which forgets that the bottom of A is one unit higher. That gives $h^2 - h + \sinh h = 16$ instead.
- Integrating $\pi y^2 \, \mathrm{d}x$ (rotation about the x-axis) rather than $\pi x^2 \, \mathrm{d}y$.
- Inverting $y = 2\sinh^{-1} x$ as $x = 2\sinh y$ or $x = \frac{1}{2}\sinh y$. The correct inverse is $x = \sinh \frac{y}{2}$.

Beyond STEP

The final equation mixes a polynomial with a hyperbolic function, so it has no closed-form solution. It has to be solved numerically, and the monotonicity argument above is exactly what guarantees that a method like bisection or Newton–Raphson finds the one physically meaningful root. First-year numerical analysis makes this precise with the intermediate value theorem and convergence rates. For the analysis behind it, see Abbott, *Understanding Analysis*.

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